



Society of Petroleum Engineers

SPE-227445-MS

The Biostratigraphy and Sedimentology of the Aruma Group: An Integrated Outcrop and Subsurface Study in Ras Al Khaimah, United Arab Emirates

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227445-MS](https://doi.org/10.2118/227445-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

The Santonian-Maastrichtian age Aruma Group was deposited in a foredeep basin created by stratigraphic loading due to compressional thrusting and obduction of the Hajar Mountains. The Aruma requires high resolution biostratigraphy with seismic guided sequence stratigraphy to identify mappable units and unconformities across the foredeep basin (Swire, 2022). Aruma Group sediments outcropping in Ras Al Khaimah, United Arab Emirates were reviewed for reservoir potential and have been correlated with the subsurface Juweiza-Qahlah reservoirs of Campanian age that outcrop in Sharjah and further to the south in Abu Dhabi along the thrust front.

The British Geological Survey completed geological mapping of outcrops in the Northern Emirates in 2006. The outcrops originally assigned to the Pabdeh Group were mapped in the absence of biostratigraphic data and there was uncertainty on the assigned age. RAKGAS, PKN ORLEN and Viridien undertook sedimentological logging and biostratigraphic analysis of eight samples from Tawi al Harf and Shaam outcrops near to RAK City to determine their affinity, age and depositional environment. The Tawi al Harf and Shaam samples were reassigned as Aruma Group after biostratigraphic analysis.

The Tawi al Harf outcrop is a series of cherty/siliceous calciturbidites, including grainstones, rudstones and conglomeratic traction flows (0.1-2m) with clasts reaching up to 10cm, separated by a series of red, purple and grey shale beds. Samples were dominated by siliceous sponge spicules and radiolaria and contained few age constrained marker taxa, with a range of Middle Campanian-Late Maastrichtian age. Deposition occurred in an outer ramp to upper bathyal setting in >200m water depth, dominantly in the deeper bathyal range. Siliceous forms were autochthonously deposited under reduced carbonate conditions, with subsequent carbonate deposition via turbidity currents. This is potentially representative of the Juweiza Formation.

The Shaam outcrop is a series of sandy-silty calciturbidite beds (0.1-1m) with hummocky cross stratification and normal grading, protruding between calcareous mudstones. Samples contain taxa that range from Early Campanian-Early Maastrichtian age. The abundance of globular planktonic foraminifera and the scarcity of larger, carinate planktonic foraminifera suggested deposition within a middle to outer

ramp setting, between 50-200m water depth. Although different than the Tawi Al Harf outcrop, it is thought that the Shaam outcrop could be assigned to a nearer shore facies of the Juweiza Formation.

Eight wells penetrating the Aruma Group were tied across Ras Al Khaimah, integrating sedimentological and biostratigraphic data obtained from outcrops and geophysical logs and biostratigraphic data from wells. Mappable units were correlated between wells and interpreted on 3D PSDM seismic data, revising the structural model by demonstrating thrusts within the Aruma. Compressional fault duplexes developed in front of the Hagab thrust, which thicken the Aruma with repeated sections in the present-day onshore subsurface area.

Key Words: Aruma, biostratigraphy, sequence stratigraphy, reservoir, sedimentology

Introduction

The compressional tectonic regime of the Late Cretaceous to Tertiary in Ras Al Khaimah, United Arab Emirates resulted in thrusting and stacking of obducted thrust sheets, causing loading, subsidence, and the formation of a foreland basin to the west of the thrust front (Swire et al., 2024). Movement on the thrust front was associated with imbrication and duplex stacking of the Late Cretaceous Aruma Group, ophiolite, and sub-ophiolite units which terminated in the Campanian (Clowser et al 1983). The foreland basin was filled with Aruma Group deposits in the Late Cretaceous and the Pabdeh Group deposits from the Palaeocene-Oligocene (Figure 1).

	Epoch		Age (Ma)	Northern UAE	Western UAE	Tectonic Interpretation
Neogene	Pliocene	Fars	5.3	Mishan Fars Anhydrites Cyclic Salts Massive Salts	Gachsaran	Late Continent - Continent Collision
	Miocene		23	Asmari		Zagros Folding and Thrusting
Palaeogene	Oligocene	Pabdeh	33.9		Asmari	Final closure of Neo- Tethys
	Eocene		56	Pabdeh	Dammam Rus	Early Continent – Continent Collision
	Palaeocene				Umm Er Radhuma	Hagab Thrust
						Musandam Culmination
Cretaceous	Maastrichtian	Aruma	66	Simsima	Simsima	Ocean – Continent Collision
	Campanian		72.1		Qahlah	
	Santonian		83.6	Fiqa	Fiqa	Semail Thrust Sheets

Figure 1—Foreland basin fill stratigraphy of the Late Cretaceous to Tertiary in the Northern Emirates and Abu Dhabi (Swire et al., 2022).

RAKGAS and PKN ORLEN are interested in oil and gas stored reservoirs within the Aruma Group. The Aruma Group requires high resolution biostratigraphy with seismic guided sequence stratigraphy to identify mappable units and unconformities across the foredeep basin (Swire et al., 2022). Biostratigraphic analysis of Cretaceous to Tertiary from nine wells (Al Khair-1X, Al Khatt-1X, Aman-1, Brooz-1X, Ghalilah-1X, Jazirat Hulaylah-1X, Najah-1X, Oryx-1, Saleh-7 and West Jiri-1X) was undertaken by Viridien (Figure 2). Sedimentological and QEMSCAN® mineralogical and petrographical analyses was also undertaken at Al Khair-1X, Aman-1, Najah-1X, Saleh-7 and West Jiri-1X wells. The British Geological Survey

The map displays the Ras Al Khaimah area with various geological features and seismic data. Key locations and features include:

- Geological Blocks:** Block 5 (central), Block 6 (east), and Block 7 (south).
- Wells:** Numerous wells are marked with yellow dots, including Saleh-7, Najah-1X, Aman-1, Jazirat, Hulaylah-1X, Brooz-1X, Oryx-1, Al Khair-1X, West-Jiri-1X, and others.
- Outcrops:** Marked with green diamonds, including Shaam, Ghalilah-1X, and Tawi Al Harf.
- City:** Ras Al Khaimah is marked with a red dot.
- Seismic Data:** The map shows 2D seismic data (thin lines) and 3D seismic data (shaded areas).
- Scale and Orientation:** A scale bar indicates 40km. A north arrow is present in the top left corner.
- Legend:**
 - Green diamond: Outcrops
 - Yellow dot: Analysed wells
 - Red dot: City
 - Thin line: 2D Seismic
 - Shaded area: 3D Seismic

Figure 2—Location of Khor Kwair, Shaam and Tawi Al Harf outcrop study areas and nine studied wells.

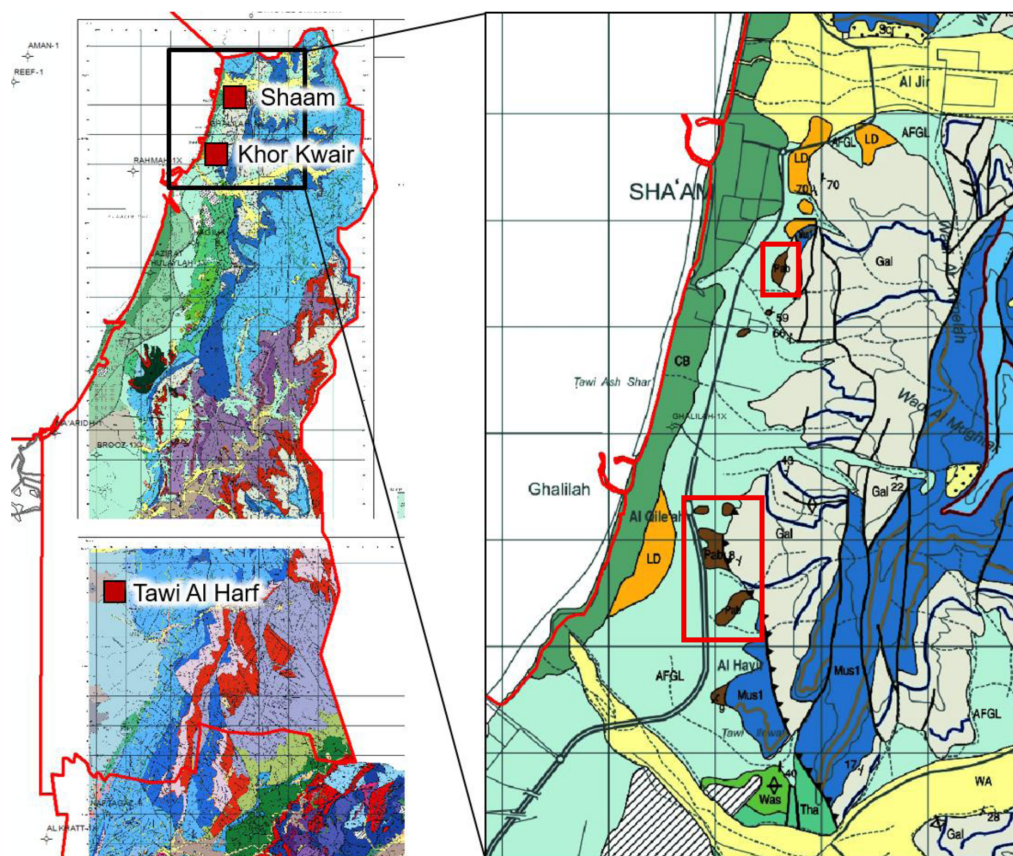


Figure 3—Location of Khor Kwair, Shaam and Tawi Al Harf study areas on the 2006 BGS Geological map. Brown areas are locations assigned to the Pabdeh Group of Tertiary age, unconformably overlying light green areas (representing Middle-Late Cretaceous Aruma Group).

Table 1—Analytical matrix of analyses and sample material / data available for each study well.

Site	Top Depth (m MD)	Bottom Depth (m MD)	Sample Type	Sedimentology			Biostratigraphy		
				QS	TS	TS Scan	M	N	P
Aman-1	1255.80	5440.70	Cuttings	48	48	48	97	95	x
Al Khair-1X	304.80	3490.00	Cuttings	44	44	44	143	65	x
Al Khatt-1X			Review				Review of Report No. 3921/1b		
Brooz-1X			Review				Review of Report No. 2903P/F		
Ghalilah-1X			Review				Review of Report No. 3135P/F		
Jazirat Hulaylah-1X			Review				Review of Report No. 3135P/F		
Oryx-1			Cuttings	-	-	-	Reported in Report No. GH279		
Najah-1X	944.90	4791.50	Cuttings	-	-	-	38	44	8
Saleh-7	1341.10	5026.20	Cuttings	-	-	-	46	51	5
West Jiri-1X	307.90	3566.20	Cuttings	59	59	59	96	98	24
Outcrop				22	22	22	23	-	-
Khor Kwair			Outcrop	14	14	14	15		
Shaam			Outcrop	3	3	3	3		
Tawi Al Harf			Outcrop	5	5	5	5		
TOTAL				173	173	173	443	353	37

Sedimentology: QS - QEMSCAN®, TS - Petrographical characterisation, TS Scan - while thin section scan

Biostratigraphy: N - Nannopaleontology, M - Micropaleontology, P - Palynology; * Al Khair-1X also incl. review of Report No. 3496P/F

Methods, Procedures, Process

All outcrops labelled as Pabdeh Group on the BGS geological map of Ras Al Khaimah were visited to determine their suitability for sedimentological logging and to obtain samples for biostratigraphic and reservoir characterization. The Khor Khwair, Shaam and Tawi Al Harf study areas were selected for logging and sample collection due to their continuous sections without significant missing sections. Samples were selected in argillaceous fine-grained beds, altered fault zone sediments and competent calciturbidite-debrite beds for determination if the outcrops were to be assigned to the Pabdeh or Aruma Group.

Reservoir potential and units were mapped between studied well locations in the Aruma Group. A biostratigraphic summary log and chart for each well was created. This included wireline logs, biostratigraphic ages, zones, sedimentological and palaeoenvironmental interpretation. Integrated QEMSCAN® mineralogical and petrographical data summary logs correlated by a biostratigraphic summary log were created for the Al Khair-1X, Aman-1, Najah-1X, Saleh-7 and West Jiri-1X wells. Age dating and zonal resolution within the Late Cretaceous was driven by nannopalaeontology, micropalaeontology and palynology and interpretation at each well was of moderate to high confidence. The nannofossil assemblages were of low diversity and at times poorly preserved, though almost always abundant. Reworking of older Late-Middle Cretaceous sediments was considered, as reworking was present in all reviewed wells and lowered confidence in interpretations. Preservation of micropalaeontological assemblages was generally very low, as was the fossil recovery of marker taxa. This meant the data generated were used as broad constraint and predominantly in palaeoenvironmental interpretation.

Results

Biostratigraphy

This study determined that outcrops at the Shaam and Tawi Al Harf locations on the BGS geological maps were incorrectly assigned to the Pabdeh Group (CGG Services UK LIMITED, 2022). Instead, Aruma Group deposits of Maastrichtian age outcrop at Shaam in Northern Ras Al Khaimah and Campanian to Maastrichtian age outcrop at Tawi Al Harf in Central Ras Al Khaimah. Aruma Group deposits were confidently reassigned in legacy well Al Khair-1X at the base of the well, having previously been assigned as Tertiary age Pabdeh Group in the absence of biostratigraphic data (Clowser, 1986, ROBERTSON RESEARCH INTERNATIONAL LIMITED, 1986). Wells Al Khatt-1X Brooz-1, Ghalilah-1X and Jazirat Hulaylah-1X were reviewed and a match to the previous interpretation was achieved (Clowser et al., 1983; ROBERTSON RESEARCH INTERNATIONAL LIMITED, 1983; ROBERTSON RESEARCH INTERNATIONAL LIMITED, 1984; ROBERTSON GROUP Plc., 1988). Well Oryx-1 was confirmed to have terminated in the Pabdeh Group (CGG Services UK LIMITED, 2023). A biostratigraphic correlation from Northern to Southern Ras Al Khaimah was achieved (Figure 4).

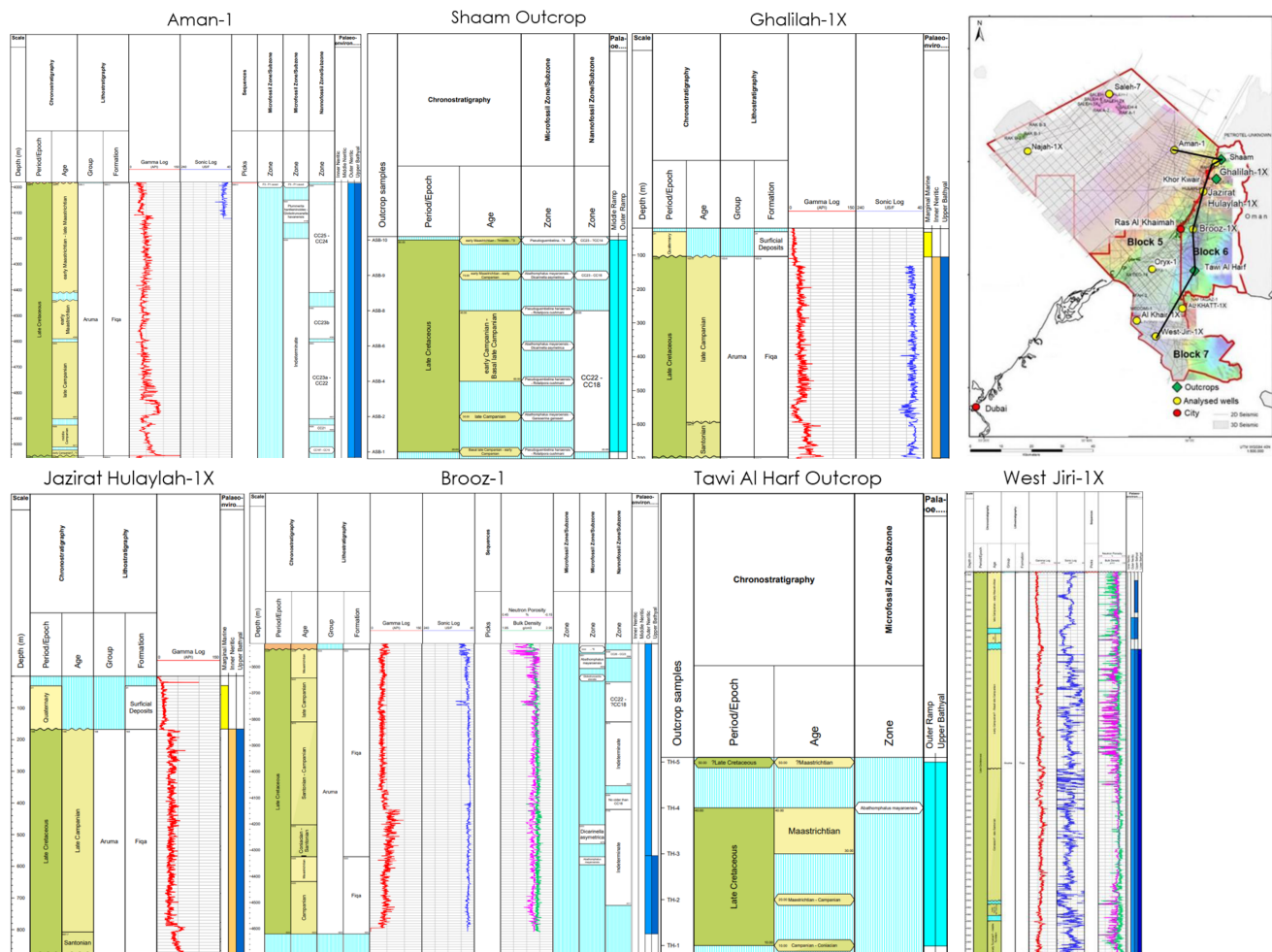


Figure 4—Aruma Group Campanian-Maastrichtian Biostratigraphic, Petrophysical and Sedimentological Correlation Panel from Northern Ras Al Khaimah to Southern Ras Al Khaimah.

The thickest and most complete series of Aruma Group sediments are found in West Jiri-1X. The well contains the only moderate confidence Turonian sediments, based on several nannopalaontological marker taxa. The taxa range no older than Turonian and rare taxa that are restricted to the Turonian. Low recovery of nannofossil marker taxa through the Late Cretaceous stratigraphy hampered the provision of a regional high-resolution age breakdown. Sediments of Santonian age were identified regionally in a number of the thicker wells, but often poorly constrained and the potential for Coniacian sediments cannot be ruled out. Caving was an issue within the Late Cretaceous and this meant that the disappearance downhole of *Micula staurophora* could not be relied on (no older than intra-Coniacian). Campanian to early Maastrichtian aged sediments were determined in all wells, although the Early Campanian could not be uniquely determined. This could indicate a regional unconformity is present, though with the current sample spacing it is hard to confirm. By the time early Campanian markers were recorded it is more likely that Santonian strata had been penetrated. The higher gamma ray log profiles would generally support this, together with an associated INC *Eiffellithus eximius*. Aman-1 contained a nannofossil marker event at 4559.9m-4590.3m – *Criboconora echinus* - that is restricted to basal early Maastrichtian sediments and can be correlated globally, for instance into sections in Spain ([Pérez-Rodríguez et al., 2012](#)) and Eastern Tanzania ([Lees, 2007](#)). This event is also seen in West Jiri-1X at 1981.2m. It is for this reason that in Aman-1 the thick section of sediments overlying this event was assigned to the late-early Maastrichtian. The sediments were lacking higher resolution age-diagnostic markers hence the ranging assignment. Therefore, Aman-1 has the most complete Palaeogene/Cretaceous boundary out of the studied wells with potentially only 3-4 Ma absent.

Late Maastrichtian sediments were absent in much of the study area, although they were identified in the Shaam outcrop and potentially in Aman-1. The studied wells incorporate a significant thickness of Early Maastrichtian to Middle Campanian aged strata. Confirmation on presence and thickness of the early Campanian is hampered, by poor recovery or fossil preservation and wide sample spacing. Depositionally, wells locations accumulated under outer neritic to upper bathyal conditions. Santonian to Coniacian sediments are present in several of the study wells and appear to overlie, potentially unconformably, Turonian-aged sediments. It is worthy of note that the Cenomanian was not identified in any of the study wells; however, this 'Middle' Cretaceous stratigraphy is assigned only a moderate level of confidence due to poor fossil recovery, very poor fossil preservation, and wider sample spacing.

Sedimentology, Mineralogy & Depositional Environment

Outcrops and studied well sections were composed of alternating fine-grained sediments of moderate to deep-water origin (claystones, siltstones, and siliceous limestones rich in planktonic foraminifera, radiolarians, and sponge spicules with frequent cherts), interbedded with fine to coarse-grained sandstone turbidites and possible debris flows.

Shaam Outcrop Area

The Shaam outcrop is a series of domed ridges with prominent hard silt-sand grade limestone beds protruding between splintered calcareous mudstones. The outcrop was logged "up-section", and limestone beds and calcareous mudstones were sampled for analysis (Figure 5). The outcrop was dated to be Maastrichtian (Late Cretaceous) in age and part of the Aruma Group.

The outcrop is 50m in length and consists of North-South striking beds that dip to the east. The base of the outcrop is a competent sandy limestone grading to a calcareous siltstone bed. 55cm of finely laminated calcareous siltstone beds pass upwards into a 45cm thick well sorted, rounded and fine grained sandy limestone with swaley cross lamination. Rare calcite cemented nodules and chert replacement were seen. The downward truncated hummocky cross laminations and inverse grading suggest that the outcrop is overturned, and the base of the outcrop is the youngest bed.

Up-section the thickness of the sandy and silty limestone beds decreases (10-20cm) but there is an increase in the frequency of these bedsets. The silty and sandy limestone beds are correlatable across ridges (100s meter scale). Converging small scale faulting (~20cm throw) offset the thin beds and made tracing their lateral extent difficult. The Shaam outcrop is heavily fractured and faulted with calcite fault rock growth fibers in variable orientations but generally indicating close to dip slip. Unsurprising in overturned beds close to a major thrust front.

The outcrop area appears to be overturned. The splintered calcareous mudstone likely represent periods of pelagic deposition that were disrupted by influx of sandy and silty calciturbidites that are compositionally variable and containing a range of mineralogical constituents indicating reworking from the hinterland.



Tawi Al Harf Outcrop Area

The Tawi Al Harf outcrop is a North-South striking ridge around 250m in length that is composed of two limbs separated by a path/road. A composite log section was created, working up section from the oldest exposed beds at the base of the outcrop (Figures 6-7).

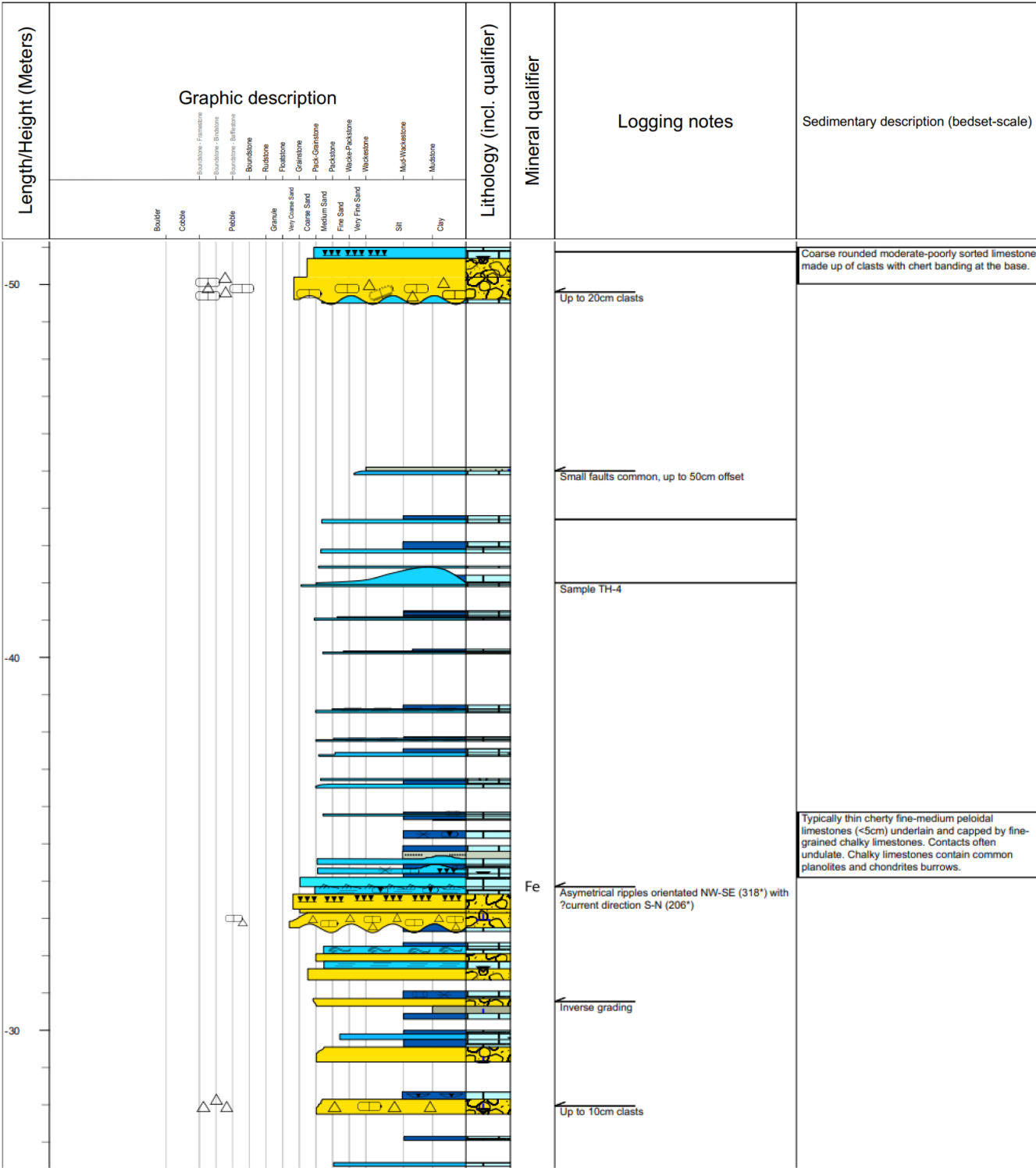


Figure 6—Tawi Al Harf Aruma Group Campanian-Maastrichtian outcrop summary log (Upper).

Locality 1 is around 50m of logged section. The base of the interval is composed of cherty/siliceous iron rich grainstone, rudstones/conglomerate beds (sample TH1) around 0.5-1m thick these beds are separated by a series of red/purple and grey shale beds (sample TH2). QEMSCAN® and petrographic analysis revealed that these are siliceous-rich radiolarian beds. Fine-grained buff coloured limestone beds around 8m into the logged section contain possible planolites and chondrites burrows. At 24m of the logged section there is the first thick (~1.8m) significant conglomerate/rudstone bed. The base is a very coarse clast supported conglomerate. No obvious orientation, sorting or grading of the clasts was observed. Clasts reach up to 10cm in diameter and are composed of limestone fragments, quartz and lithics. The matrix is calcite and silica cemented. The base is overlain by fine sand grade cross laminated limestones with partial chert replacement. The bed is capped by a coarse- to very coarse-conglomerate with similar composition.

The bed passes sharply (undulating inclined contact) into a fine sand grade limestone with swaley cross laminations and rare chert nodules. At 34m on the logged section is the next major conglomerate bedset. The top of the unit is covered in asymmetrical ripples. Above this is a series of thin cherty fine-medium sand grade peloidal limestones (sample TH4) which are typically <5cm in thickness. These are underlain and capped by fine-grained chalky limestones, with contacts often undulating. The chalky limestones contain common planolites and chondrites burrows. At 50m of logged section is the last conglomerate bedset. Logging was stopped here due to the rubbled nature of the outcrop above this point, but also because of the proximity to a tight fold in competent limestone beds. This suggested the potential for significant deformation (folding, bed parallel slip, thrusting) in poorly exposed less competent material: possible thickening/repetition of the section was impossible to estimate. The measured section at this outcrop was a maximum for a complete section capable to be logged. Sample TH5 was taken away from the two logged sections as a feature of interest. The bed is an oolitic grainstone that had been unseen whilst logging. QEMSCAN® and petrographic analysis revealed that the sample is replaced by silica. The siliceous mudstones at the base of the section potentially represent periods of more anoxic waters while the buff-coloured burrowed limestone beds may indicate more oxygenated waters and pelagic deposition. The conglomerate beds are likely traction flows at the base of calciturbidites. The TH5 ooid grainstone bed probably represents re-sedimentation of shallow-water carbonate deposits into the foredeep basin. The outcrop itself is structurally complicated but the composite logs gave an idea of Late Cretaceous deep water depositional systems in the Aruma Group.

Middle Coniacian - Middle Campanian. Cuttings data from this interval was present in West Jiri-1X, Al Khair-1X, and Aman-1 wells. However, in West Jiri-1X cuttings are compositionally and mineralogically distinct and between basal Late Campanian to potentially Early Campanian age. The cuttings are dominated by siliceous/radiolarian-rich mudstones and mixed kaolinite/illite mudstone chips, with variations between the two possibly indicating cyclicity in water depth and/or dilution of the prevailing background sediment character by increased terrigenous/siliceous mudstone input. Notable other features unique to this well also include coincident with the Late Santonian to Coniacian / Basal Late Campanian-Early Campanian boundary the presence of muddy debrite and disaggregated sandstone chips, which may mark increased turbidite deposition during this interval in the proximity of West Jiri-1X. Alternatively, they potentially suggest localised and more regional remobilisation events e.g. mass transport complexes. Siliceous mudstone deposition is noted to continue after this interval, however, notably within the latest Early Campanian appears ‘diluted’ by increased calcareous mudstones and peloidal/oolitic limestone chips, reflecting possible subtle variations in water depth, based on the fluctuations of calcareous mudstone fragments, and as well as periodic input of shallow marine carbonates into the basin via possible calciturbidites.

Late Campanian – Late Maastrichtian. Within the overlying Late Campanian – Late Maastrichtian deposits, cuttings in all wells are dominated by planktonic foraminifera-rich mudstones and scarcer illitic mudstone chips. This suggests that any possible tectonism captured in West Jiri-1X ceased and/or is not recorded in any of the wells, with the resumption back to deepwater (outer neritic to upper bathyal)

mudstone, with limited terrigenous and/or reworked shallow marine input suggesting an overall quiescent sediment-starved depositional setting. The proximity of the West Jiri-1X well to the Hagab Thrust zone during this interval is possibly decisive in terms of its depositional character and may represent a localised source of the siliceous mudstone and sandstone material that is absent in the Al Khair-1X and Aman-1 wells. Outcrop evidence at both Shaam and Tawi Al Harf localities are generally consistent with proposed outer neritic to upper bathyal setting, with the presence of calciturbidites and conglomeratic/traction flow beds indicating variable scale turbidite input, and at Tawi Al Harf possible increased proximity to the feeder source. More information on each of these outcrop localities is presented in Chapter 5 below. Deposition of the Late Cretaceous is interpreted to have taken place in an outer neritic to upper bathyal setting deepening towards the Palaeogene/Cretaceous boundary and onset of the deeper water Pabdeh Formation deposition. In West Jiri-1X the early Maastrichtian sediments (which are the last Cretaceous sediments present in the well due to the unconformity) are deposited under upper bathyal depths.

Seismic Correlation

The well and outcrop locations were correlated to generate a correlation panel highlighting several thrusts in the Pabdeh to Aruma Group from Northern Ras Al Khaimah to Southern Ras Al Khaimah (Figure 8). The biostratigraphic, sedimentological and petrophysical correlation panel of study wells and outcrops was integrated with 3D PSTM seismic data from offshore and onshore Ras Al Khaimah, a 2D seismic composite line east of Ras Al Khaimah and outcrop studies. Structural transects from North-South and West-East Ras Al Khaimah were created to characterize the Aruma section in the subsurface (Figure 9).

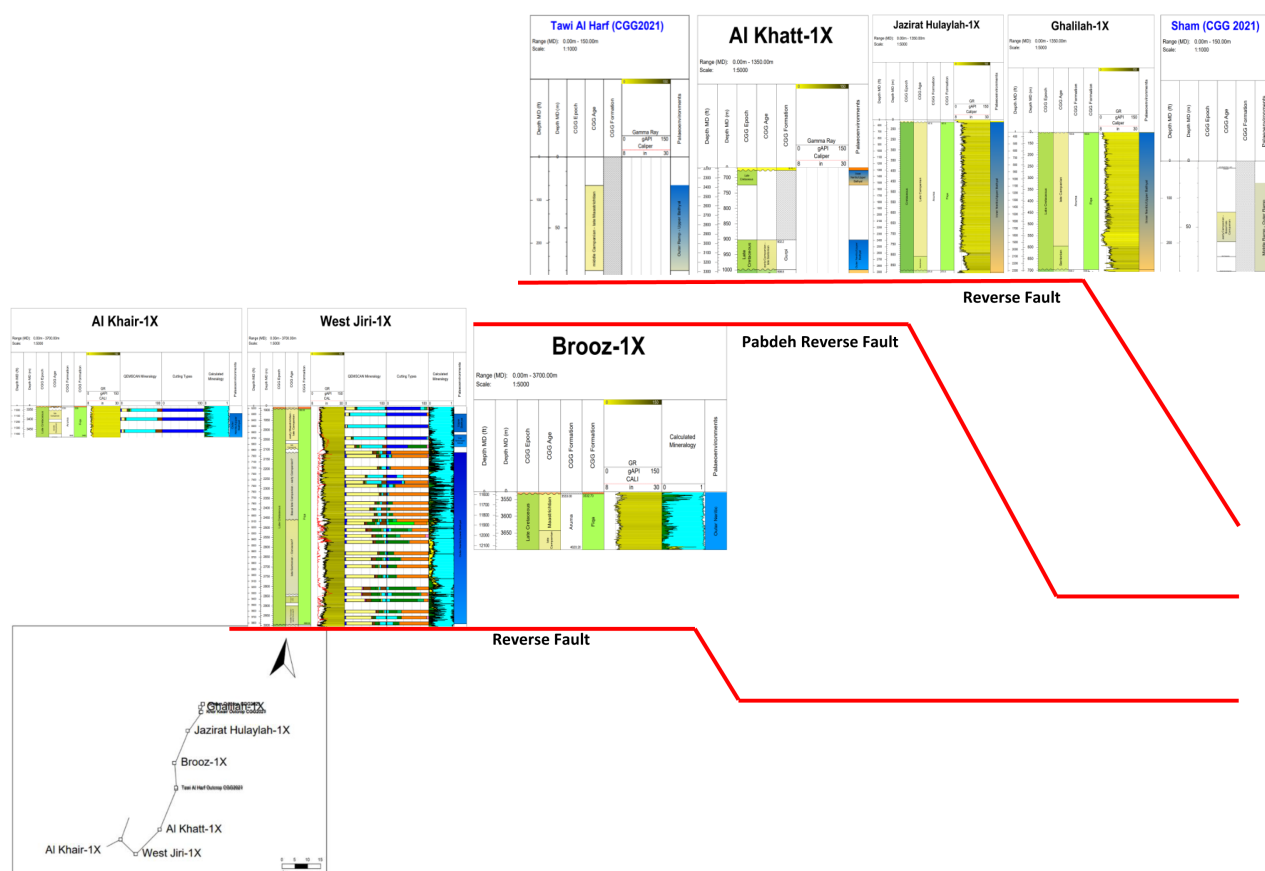
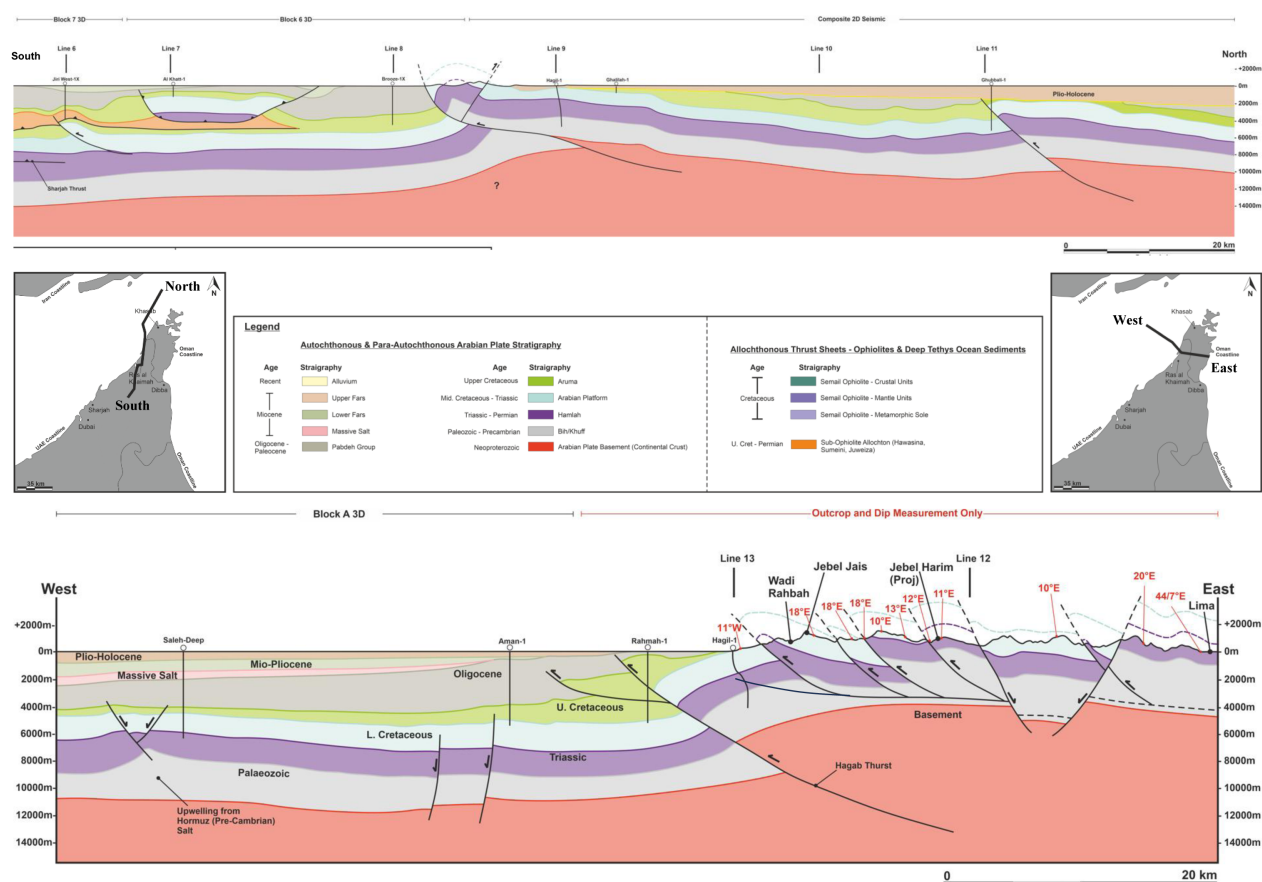


Figure 8—Aruma Group Campanian-Maastrichtian Biostratigraphic, Petrophysical, Chemostratigraphic and Sedimentological Correlation Panel from Northern Ras Al Khaimah to Southern Ras Al Khaimah. Thrust locations within the sections are highlighted.



Conclusions

The Tawi Al Harf and Shaam outcrops were confirmed to be Late Cretaceous in age (from early Campanian to late Maastrichtian) and part of the Aruma Group. The Shaam outcrop was likely deposited in 50-200m water depth while the Tawi Al Harf outcrop was likely deposited in greater than 200m water depth. Both areas show large sections of fine-grained pelagic beds interspersed with coarser debris flows. The Shaam outcrop contained finer calciturbidite beds with common hummocky cross stratification and normal grading. Bedsets are typically between 0.5-1m. Tawi Al Harf contains impressive 1-2m thick conglomeratic traction flows with clasts reaching up to 10cm. The bedsets contain resedimented shallow water limestones and formed a series of calciturbidites. This study has opened up a new reservoir target in Ras Al Khaimah for the calciturbidite and conglomeratic traction flows of the Campanian-Maastrichtian.

A high resolution stratigraphic, mineralogical and depositional model of the Cretaceous Aruma Group was achieved for the Northern U.A.E. The integration of field studies, well cuttings and core analysis with seismic attribute interpretation generated a correlation of outcrops to the subsurface in Ras Al Khaimah. Channelisation and basin floor fan development is a key event in the development of the basin. Integration between detailed biostratigraphy and mineralogical analyses reveals a robust sequence stratigraphic framework and correlation of study wells. The depositional facies map and interpretations are based on limited data points from wells, outcrops, and structural trends. To refine this approach and improve regional understanding and facies/reservoir quality trends, recommendations for future study are as follows:

- Integration of additional well data, plus combined wireline log interpretation notably over the over early deep water turbidite deposits to help refine the depositional facies interpretation and reservoir predictability in areas that have limited well control.

- Further refinement of the stratigraphic interpretation through the integration of regional PSDM seismic data and possible additional outcrop localities to assess calciturbidite/debrite geometries and possible sediment fairways.
- Improved reservoir quality understanding and reservoir quality predictability of key reservoir zones through detailed core description, image log interpretation and petrographical analyses of core/SWCs (or incorporation of existing data and wells into this study).
- Improved sampling density and targeted sampling over the 'Mid'/Late/Early Cretaceous boundary, which is currently poorly understood, to define a more detailed and high-resolution framework for the Aruma Group

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